Source Regions and Atmospheric Transport during Summertime Ozone Peak Events in the Northern Colorado Front Range

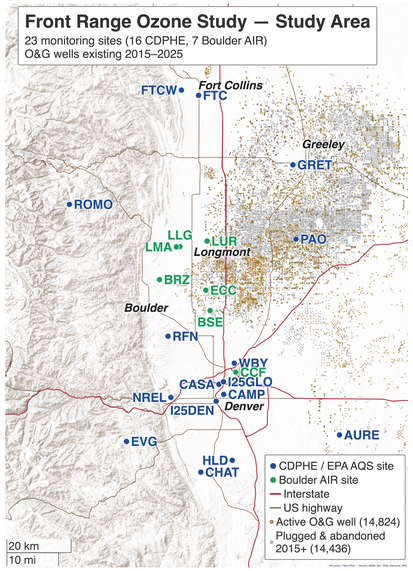
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Abstract

Constraining the geographic origins of ozone and its precursors remains an important challenge for air-quality modeling and source-attribution studies. This study funded by the Colorado Air Quality Enterprise combines continuous observations of ozone, nitrogen oxides (NOx), and volatile organic compounds (VOCs) from 23 monitoring sites across the Denver Metro/Northern Colorado Front Range (DM-NCFR) with NOAA HYSPLIT back trajectories and Potential Source Contribution Function (PSCF) analysis to identify source regions associated with elevated ozone transport. To support the analysis, 285,103 twelve-hour back trajectories were generated using 3-km High-Resolution Rapid Refresh meteorological data, spanning eleven ozone seasons (2015–2025). Elevated pollutant conditions were identified using station-specific percentile thresholds and paired with back trajectories to construct PSCF source maps. Initial analyses focused on data from local government-funded monitoring stations at Union Reservoir in Longmont and the Erie Community Center.

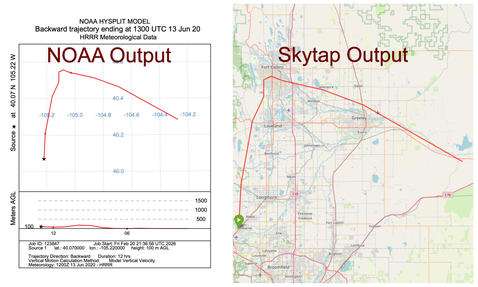
Preliminary results suggest that elevated ozone occurrences are relatively uniform along the Front Range and reflect the mixing of NOx transported from the south with VOCs transported from the northeast, with both trapped against the eastern slope of the Rocky Mountains. High total-carbon VOC values point to a dominant source region northeast of Longmont that closely matches the high density of oil and gas wells in the area. In contrast, high-NOx source regions are more closely associated with urbanized areas in the western and southwestern Denver metro area. These findings suggest that transport of VOC-rich air masses from areas of dense oil and gas operations to the northeast makes an important contribution to elevated ozone episodes in the DM-NCFR. The VOC-rich (high VOC/NO_x ratio) Denver Julesburg Basin appears to be a more efficient ozone-producing region than the higher NO_x (lower VOC/NO_x ratio) Denver Metro area. Ongoing analyses will expand PSCF evaluation to additional monitoring sites and develop multi-site source attribution methods to test whether independent sites consistently identify the same source regions. Agreement across sites will turn broad directional signals into more defensible regional source attribution.



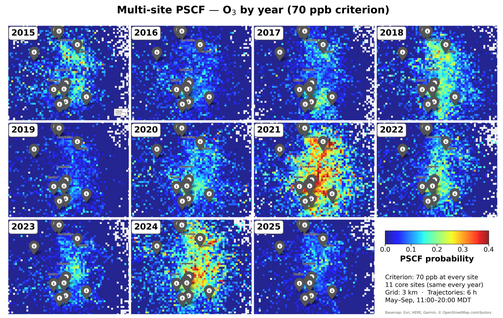
1 Study area. The 23 monitoring sites (blue: 16 CDPHE; green: 7 Boulder AIR) span the Denver Metro / Northern Colorado Front Range. Orange dots are the 14,824 active oil and gas wells; gray dots are wells plugged and abandoned since 2015. Well development is concentrated east and northeast of Longmont and Erie, around Greeley. Union Reservoir (LUR) and Erie Community Center (ECC) are the two sites analyzed first.

Video: How PSCF Works

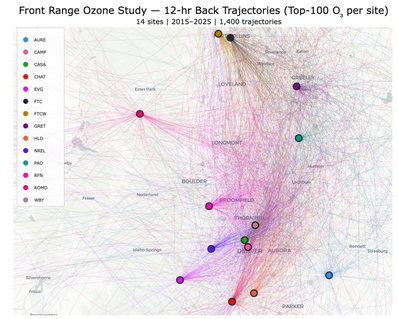
A conceptual walkthrough of the Potential Source Contribution Function, from back trajectories to source-region maps. Scan to watch (1 min).



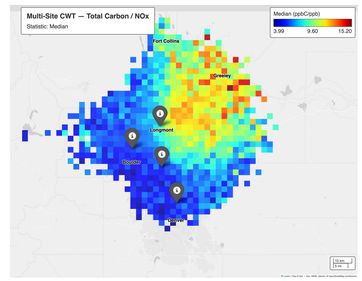
2 Tracing air backward. Air arriving near Gunbarrel (green marker) on 13 June 2020, traced back 12 hours to east of Greeley. **Left:** NOAA's HYSPLIT output. **Right:** the same trajectory from Skytap, our custom program that generated all 285,103 trajectories in parallel. We tested Skytap's output against NOAA's, and the paths match.



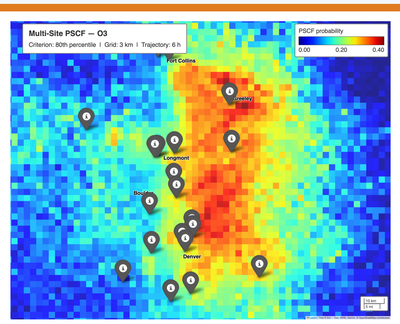
4 Year to year. Ozone PSCF for each year, 2015–2025, using a fixed 70 ppb cutoff at the 11 sites that ran all 11 years (May–Sep, 11:00–20:00 MDT). There is a relatively large year-to-year variability in peak surface ozone, but the ozone source region is relatively consistent.



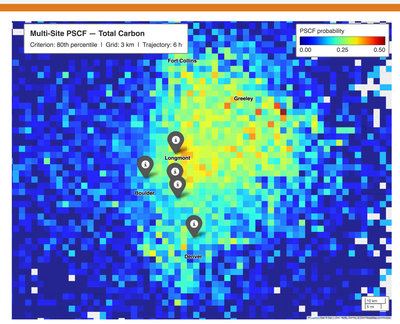
3 High-ozone air, all sites. 12-hour back trajectories for the 100 highest-ozone hours at each of the 14 sites (1,400 trajectories, 2015–2025); each color is one site. Across most sites, high-ozone air tends to arrive from the north and east.



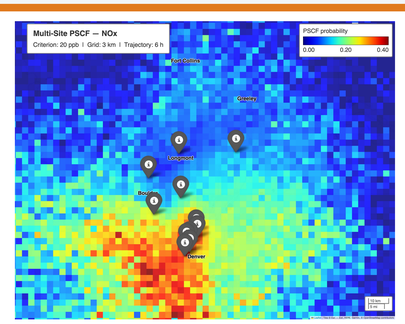
5 VOC-rich air comes from the northeast. Multi-site concentration weighted trajectory (CWT) map: each trajectory is tagged with the total VOCs carbon / NOx ratio (ppbC/ppb) measured when it arrived, and each 3-km cell shows the median ratio of the trajectories that passed over it. Ratios peak around Greeley and taper off to the south and west.



6 Where high-ozone air comes from. Multi-site PSCF (6-hour trajectories; high ozone = above each site's own 80th percentile). The hot spot stretches from Fort Collins to southeast of Denver, spanning both the oil and gas fields (Figure 7) and the urban NOx sources (Figure 8).



7 VOCs come from the oil and gas field. Multi-site PSCF of total VOCs carbon (ppbC of 16 C2–C8 hydrocarbons, ethane through xylenes; above each site's 80th percentile). High values extend northeast from Longmont and closely track the densest oil and gas development in the Denver-Julesburg Basin (Figure 1).



8 NOx originates in the Denver Metro urban core. Multi-site PSCF of NOx using a fixed 20 ppb criterion, which isolates the source regions of the highest regional concentrations. The dominant source regions lie south of Boulder and across southwest Denver, overlapping the southern portion of the ozone hot spot in Figure 6.

Funded by the **Colorado Air Quality Enterprise**. Results are preliminary and subject to change. Trajectories: NOAA HYSPLIT with 3-km HRRR meteorology.
Contact: **gabrielgreenbergwork@gmail.com**
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